

Twin Pregnancy 06. Fertility, Reduction, and Counseling

Companion reading handout for simultaneous listening.

Audio course on twin pregnancy. Episode six. Fertility treatment, fetal reduction, and counseling.

Source base

This episode is based on Novak chapter 36 on infertility, Gabbe chapter 32 on multiple gestations, and the Israeli position paper on fetal reduction.

Opening frame

Let us begin with the deepest idea.

The safest way to manage iatrogenic multiple pregnancy is to prevent it before it exists. Once a high-order multiple pregnancy or a discordant fetal anomaly exists, counseling becomes a risk-balancing exercise rather than a simple recommendation.

Start with fertility treatment

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Assisted reproduction and ovulation induction increased the frequency of twin and higher-order pregnancies. Ovulation induction can recruit more than one follicle. Superovulation intentionally increases the number of available oocytes. Embryo transfer can place more than one embryo into the uterus.

For residents, the key concept is that the desire for pregnancy can conflict with the desire for a safe pregnancy. Some patients view twins as an efficient or even desired outcome, especially when fertility treatment is expensive or emotionally exhausting. But medically, twin pregnancy increases preterm birth, low birth weight, neonatal morbidity, hypertensive disease, hemorrhage, cesarean delivery, and long-term neurodevelopmental risk.

That is why elective single embryo transfer is not simply an infertility-lab policy. It is obstetric prevention.

Novak emphasizes that multiple gestation is one of the major risks of assisted reproductive technology, and that adherence to embryo-transfer limits and improved implantation rates have reduced multiple live birth rates. Even with single embryo

transfer, monozygotic twinning can occur, and the rate after in vitro fertilization is higher than the spontaneous background rate.

Now move from prevention to established high-order pregnancy

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Fetal reduction means intentionally reducing the number of fetuses to improve the outcome for the remaining fetus or fetuses and the mother. The Israeli position paper separates two situations.

The first is multifetal pregnancy reduction, or MFPR, usually in triplets or higher-order multiples, where the goal is to reduce the risks that come from fetal number itself.

The second is selective reduction, usually when one fetus has a major anomaly, chromosomal disease, genetic disorder, severe infection, or a dangerous monochorionic complication.

These are ethically and emotionally different conversations, even if the procedure may look similar technically.

In high-order multiple pregnancy, reduction from triplets to twins can prolong pregnancy and reduce early preterm birth and neonatal mortality. Reduction from triplets to a singleton may reduce prematurity even further, but it also changes the parental goal from preserving twins to a singleton outcome. Counseling must make the tradeoff explicit.

The Israeli paper states that the option of fetal reduction should be presented to every couple with triplet pregnancy or more. Presenting the option is not coercion. It is informed consent.

Before reduction, chorionicity must be diagnosed. This is critical.

In dichorionic or trichorionic pregnancies, potassium chloride injection into the fetal heart or chest can stop the selected fetus because the fetal circulations are separate. In monochorionic twins, potassium chloride injection is dangerous to the co-twin because shared vascular anastomoses can transmit the drug or cause acute hemodynamic injury.

Therefore, monochorionic selective reduction usually requires cord occlusion or vascular ablation: radiofrequency ablation, bipolar cord coagulation, laser, or fetoscopic approaches. The purpose is to interrupt blood flow to the affected fetus without causing back-bleeding or toxic transfer through placental communications.

This is the second major counseling anchor. In monochorionic pregnancy, you do not only reduce a fetus. You must separate circulations.

The risk to the co-twin is substantial. The Israeli fetal-reduction paper notes that monochorionic reduction methods are associated with a significant risk of losing the other twin, depending on method and gestational age. This must be said clearly before the procedure.

Now prenatal diagnosis before reduction

Now prenatal diagnosis before reduction.

Reduction around 11 to 14 weeks can allow nuchal translucency assessment and sometimes early anatomic assessment. NIPS may be possible in twin and triplet pregnancies, but abnormal results do not identify which fetus is affected, and after reduction or vanishing twin, NIPS becomes problematic. CVS before reduction can clarify fetal karyotype in selected cases and does not necessarily add loss risk when performed by experienced operators, but it is not mandatory for every case.

The principle is to get enough information to reduce the right risk without delaying so long that the pregnancy enters a more dangerous gestational window.

Now selective reduction for fetal anomaly

Now selective reduction for fetal anomaly.

In dichorionic twins, if one fetus has a severe anomaly and the other appears normal, selective reduction can be considered. Timing depends on when the diagnosis is made, what confirmatory testing is needed, gestational age, viability, and the risk that the affected fetus could be born alive before reduction if waiting too long.

At the border of viability, the decision can be especially difficult. Earlier reduction may reduce the chance of delivering an affected liveborn infant, but if the procedure triggers labor, it may produce extreme prematurity for the healthy co-twin. Delayed reduction may reduce prematurity injury if labor follows the procedure, but it increases the chance that spontaneous labor occurs before the procedure.

This is not a technical dilemma only. It is a values dilemma, a neonatal-risk dilemma, and a procedural-risk dilemma.

Now rare configurations

Now rare configurations.

Triplet pregnancy with a monochorionic pair and a separate singleton is particularly complex. If reducing to twins, which pregnancy remains matters. Keeping the monochorionic pair preserves the vascular-risk pregnancy. Keeping the singleton and reducing the monochorionic pair may reduce monochorionic complications but

changes the fetal-number goal. The Israeli paper generally recommends reducing the monochorionic twins and leaving the separate fetus in some such configurations, while also acknowledging patient preference and individualized counseling.

Pause and summarize.

Fertility treatment can create twin and high-order pregnancy, so prevention through careful ovulation induction and single embryo transfer is obstetric risk reduction. Once high-order multiple pregnancy exists, fetal reduction is counseling about competing risks. Chorionicity determines technique. Dichorionic reduction can use fetal injection. Monochorionic reduction requires vascular occlusion because the circulations communicate.

Now final synthesis

Now final synthesis.

Key point

First, the prevention of iatrogenic multiple pregnancy belongs in obstetric thinking, not only infertility practice.

Second, fetal reduction has two broad indications: reducing fetal number in high-order pregnancy and selectively reducing an affected fetus.

Third, chorionicity determines procedural safety. Monochorionic pregnancies require cord or vascular occlusion strategies, not simple intracardiac potassium chloride.

Fourth, counseling must be honest about prematurity, procedure-related loss, co-twin risk, timing, and the couple's goals.

End of episode six.